

Networks Decide for Us

When homophily governs the criticality of social adoption

Aníbal Olivera M.

PhD(c) in Social Complexity Sciences, CICS

with Jorge Fábrega & Thomas W. Valente

The question

- Collective behaviors (innovations, mobilizations) spread as **adoption cascades** on social networks.
- Most models use *synthetic* topologies and treat agents as cognition-free relays [1].
- Real networks are shaped by **multidimensional homophily** — similar people cluster [2].

Conditional on *fixed individual propensities*, can the **shape of more realistic networks alone** change the outcome? Can the structure itself be an **autonomous force**, deciding not just *how much* spreads, but *how* it spreads?

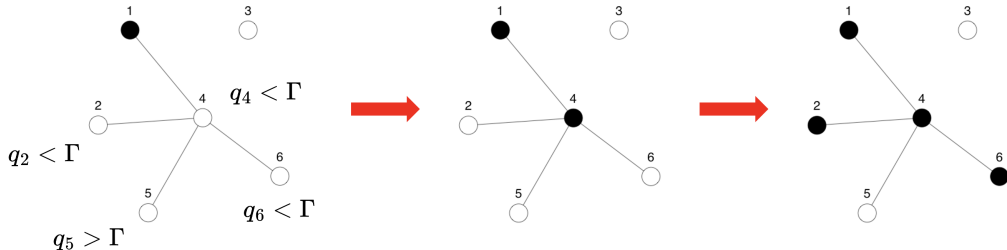
The model

The model — 1) Rational Choice

- Following the literature, the attractiveness of the innovation is its *Intrinsic Utility Level* (IUL), $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* (MUR), $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff

$$q_i \leq \Gamma \implies a_i = 1$$

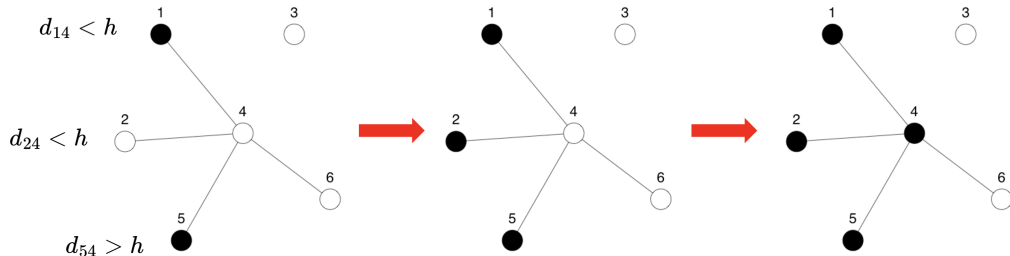


Everyone with $q_i \leq \Gamma$ adopts, regardless of neighbors: **deterministic** and **topology-blind**.

The model — 2) Selective Social Influence

If intrinsic adoption fails, an agent may still adopt by peer contagion if $\tau_i \leq \tilde{E}_i$ [5,6] — but only *influential* peers, those socially *close* in Blau space [7], count:

$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

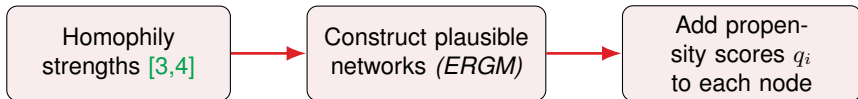


$\tau_4 = 0.5$: close adopters $d_{14}, d_{24} < MSP$ count; the distant one $d_{54} > MSP$ does *not* $\Rightarrow$ node 4 ignites from similar peers only.

MSP (Maximum Social Proximity) sets how far influence reaches; this channel is **stochastic** and **topology-sensitive**

Imputing structure & running the model

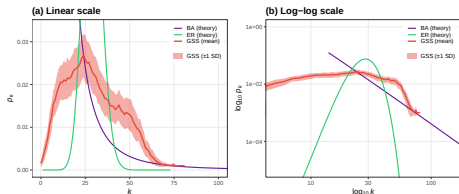
Imputing a network onto a survey



Tie probability is a function of distance in Blau space (age, sex, education, race, religion):

$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \beta^\top \mathbf{d}_{ij}$$

$\hat{\beta}$ comes from GSS confidant data [4]; propensities q_i come from real survey answers (innovation, collective action).



Right-skewed degree structure that ER / BA models fail to match jointly.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [9], over plausible (GSS) vs. **degree-preserving randomized** (GSS-DP) topologies.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

Total: $41 \times 13 \times 4 \times 4 \times 5 \times (\text{runs}) \approx 4 \times 10^6$ **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}} \mathbb{I}_{\text{DP}} + s(\text{IUL}, \text{MSP})$$

A statistical-physics reading

Mapping social adoption to a ferromagnet

Ising system	NDFU
Spin $s_i \in \{0, 1\}$	$\leftrightarrow$ Adoption state a_i
Magnetization M	$\leftrightarrow$ Adoption $\Phi = N_{\text{ad}}/N$
External field H	$\leftrightarrow$ IUL Γ (rational bias)
Temperature T	$\leftrightarrow$ MSP h (social agitation)
Exchange J_{ij}	$\leftrightarrow$ Homophily / clustering
Susceptibility χ	$\leftrightarrow$ $\text{Var}(\Phi)$ (FDT)

Pseudo-Hamiltonian:

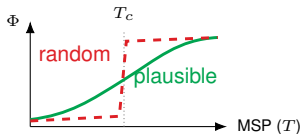
$$\mathcal{H} \approx - \sum_{\langle i,j \rangle} w_{ij}(h) s_i s_j - \Gamma \sum_i s_i$$

Low MSP $\rightarrow$ rigid echo chambers (frozen domains).

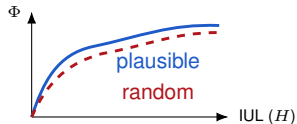
High MSP $\rightarrow$ boundaries “melt”, global cascades possible.

Susceptibility diverges near criticality [8]: $\chi = \text{Var}(\Phi) \propto |MSP - MSP_c|^{-\gamma}$

Two paths through the critical point



Path 2 — vary T : continuous (plausible, $\gamma \approx 1$) vs. discontinuous (random, $\gamma \approx 4$).



Path 1 — vary H at $T = T_c$: same critical isotherm, $\delta \approx 3$ for both.

Path 2 — vary MSP (T) at fixed IUL

- continuous **2nd-order** transition; exponent γ
- drives the **social channel** $\Rightarrow$ **topology-sensitive**
- **only the plausible network reaches $\gamma \approx 1$**

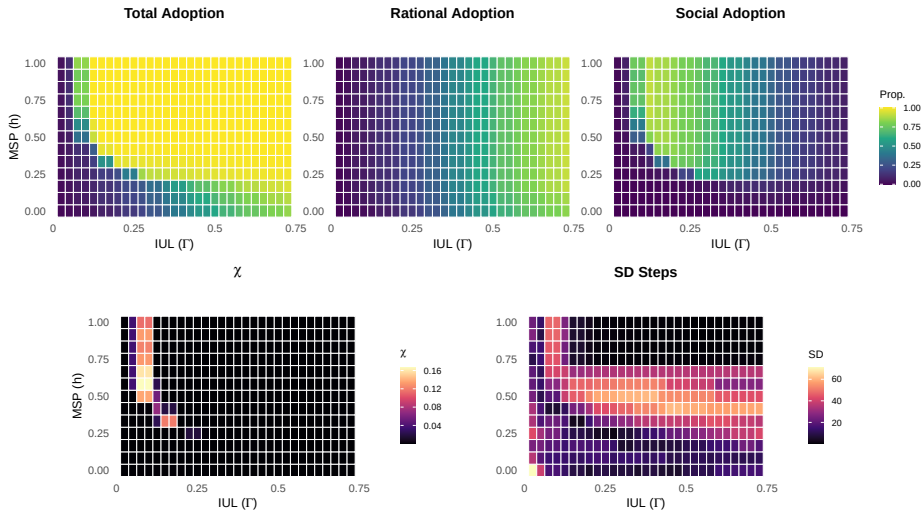
Path 1 — vary IUL (H) at fixed MSP

- field-direction response; exponent δ
- drives the **rational channel** $\Rightarrow$ **topology-blind**
- **both** networks give the same $\delta \approx 3$

Results

Result 1 — Criticality signatures (dynamics)

Emergence of Adoption and Phase Transition Signatures



Result 2 — A structural premium (volume)

BAM isolates the marginal effect of destroying empirical structure:

$$\beta_{DP} = -0.850^{***} \text{ (OR} = 0.43\text{)}$$

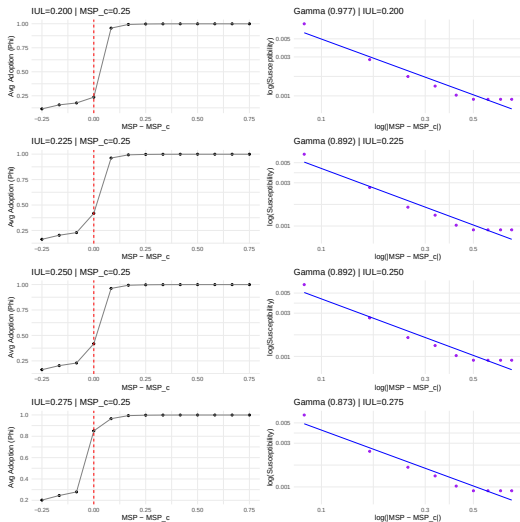
⇒ randomization cuts adoption odds by
~**57%**

Under *identical* propensities, homophilic clustering lets social reinforcement build inside **local echo chambers** — even for unattractive innovations ($\Gamma < 0.25$).

Total adoption (GSS)	Estimate	OR
(Intercept)	6.534***	—
τ_{mean}	-6.944***	—
τ_{sd}	5.942***	—
<i>Seeding (ref: Random)</i>		
Central	0.120***	1.127
Closeness	0.119***	1.126
Eigenvector	0.118***	1.125
Marginal	-0.067***	0.935
<i>Topology (ref: Plausible)</i>		
Deg.-pres. random	-0.850***	0.427
Deviance explained: 82.2% $N = 4,093,440$		

*** $p < 0.001$. OR = $\exp(\beta)$.

Result 3 — Universality is exclusive to plausible networks



IUL	Plausible γ	Randomized γ
0.200	0.98	3.71
0.225	0.89	3.67
0.250	0.89	3.67
0.275	0.87	4.67
mean	0.91	3.93

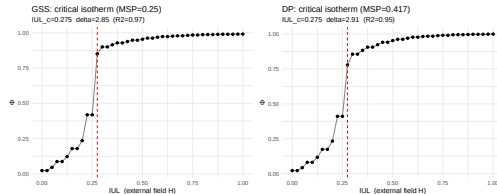
- **Only** the plausible (homophilic) network reaches $\gamma \approx 1$ — *Mean-Field universality*, a continuous transition.
- Randomization $\Rightarrow \gamma \approx 4$: no power law, **discontinuous** regime shift.
- Separation $\approx 4.3\times$.

Why only the MSP axis discriminates

Two control parameters, two channels:

- **Path 2 (MSP, γ)**: the *social* channel. $\gamma_{\text{GSS}} \approx 1$ vs. $\gamma_{\text{DP}} \approx 4$ — **discriminates**.
- **Path 1 (IUL, δ)**: the *rational* channel ($q_i \leq \Gamma$ is node-local). $\delta_{\text{GSS}} \approx 2.85$, $\delta_{\text{DP}} \approx 2.91$ — **both $\approx \text{MF } \delta = 3$** .

The field response reflects the *MUR distribution*, shared by both topologies. So “universality” is **not** generic — it is a fingerprint of empirically homophilic structure.



Critical isotherms $\Phi(\text{IUL})$: same $\delta \approx 3$ for both.
(*exploratory*)

The mean-field paradox — and its resolution

Puzzle

Mean-field $\gamma = 1$ usually needs a *well-mixed* system. We recover it in a *strongly clustered, homophilic* network.

Resolution

Homophilic clusters behave as quasi-isolated subsystems coupled by *weak, similarity-selective bridges*. As $MSP \rightarrow MSP_c$ these bridges activate together, synchronizing clusters into one coherent order parameter — *indistinguishable* from mean-field behavior. Randomization (DP) destroys this and yields discontinuity instead.

Homophily gives the system a “**sociological viscosity**”: tipping points arrive *with measurable early warning* (rising variance, slow recovery) instead of detonating blind.

Conclusions

Conclusions

Networks decide for us — and the answer to both opening questions is *yes*.

Q1. Can the shape of realistic networks *alone* change the outcome?

- **Volume:** empirical homophily yields a **structural premium** (+57% odds).
- When society is not too restrictive (high MSP), **highly attractive and unattractive innovations reach nearly the same adoption** — purely through social influence.

Q2. Is structure an autonomous force deciding *how* it spreads?

- **Dynamics:** homophily makes tipping points **continuous & predictable** ($\gamma \approx 1$); randomization makes them **abrupt & volatile** ($\gamma \approx 4$).
- Universality is **exclusive** to plausible networks; the rational (IUL) axis is topology-blind ($\delta \approx 3$).

Macroscopic social change is not the linear sum of individual preferences — it is governed by the structure of our ties.

References

1. Goldberg, A. (2021). Associative Diffusion and the Pitfalls of Structural Reductionism. *Am. Sociol. Rev.* 86(6).
2. McPherson, M., Smith-Lovin, L., & Cook, J. (2001). Birds of a Feather: Homophily in Social Networks. *Annu. Rev. Sociol.* 27.
3. McPherson, M., & Smith, J. A. (2019). Network Effects in Blau Space: Imputing Social Context from Survey Data. *Socius* 5.
4. Smith, J. A., McPherson, M., & Smith-Lovin, L. (2014). Social Distance in the United States, 1985–2004. *Am. Sociol. Rev.* 79(3).
5. Granovetter, M. (1978). Threshold Models of Collective Behavior. *Am. J. Sociol.* 83(6).
6. Valente, T. W. (1996). Social Network Thresholds in the Diffusion of Innovations. *Soc. Networks* 18(1).
7. Centola, D. (2011). An Experimental Study of Homophily in the Adoption of Health Behavior. *Science* 334.
8. Sethna, J. P. (2021). *Statistical Mechanics: Entropy, Order Parameters, and Complexity* (2nd ed.). Oxford UP.
9. Vega Yon, G., Olivera M., A., & Valente, T. (2025). *netdiffuseR: Analysis of Diffusion and Contagion Processes on Networks*. *Zenodo*.

Thanks!



Code & data: github.com/aoliveram/Networks-Decide-for-Us

an.oliveram@udd.cl